Git: The Basics

C2SM Git Courses · 26 March 2026

Annika Lauber, Mikael Stellio, Michael Jähn

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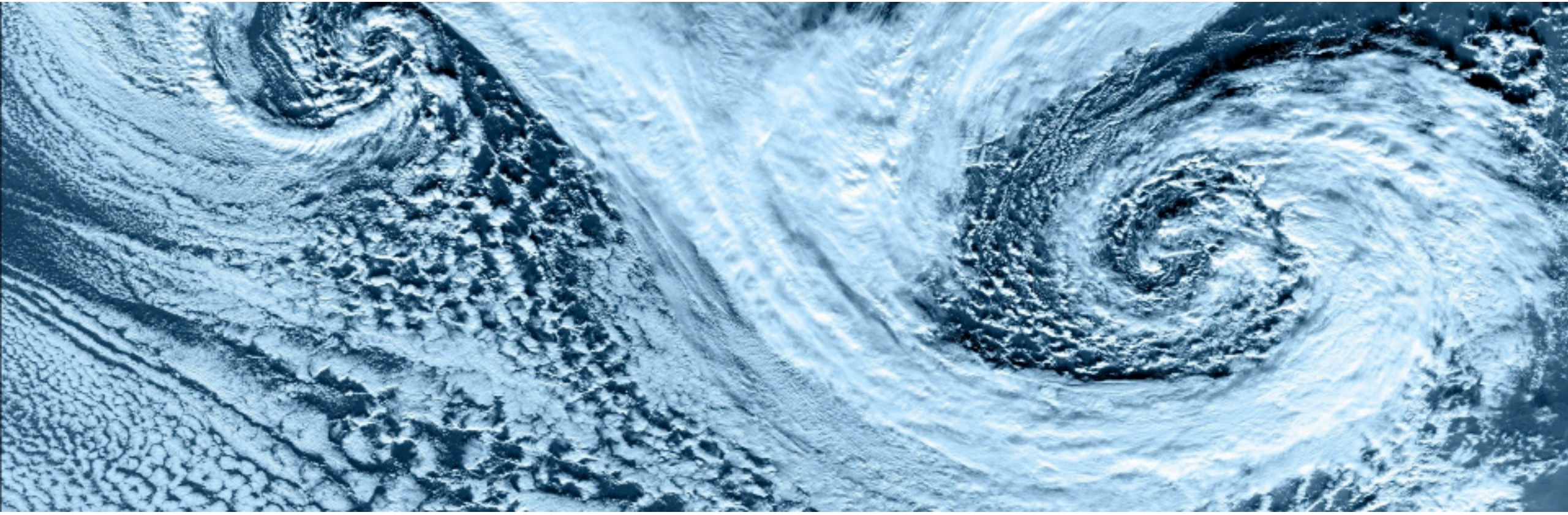

Schedule

- **9:30 - 10:30** Introduction to Git
- **10:30 - 11:00** Git Branches
- **11:00 - 11:15** Coffee Break
- **11:15 - 12:00** Merge, Restores, Delete, .gitignore
- **12:00 - 13:00** Lunch
- **13:00 - 13:30** Remote Repositories
- **13:30 - 14:00** Merge Conflicts
- **14:00 - 15:00** Repository Managers

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Part 1:

Introduction to Git

Why Use a Version Control System?

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- Tracks file changes in a documented way
  - Restore previous versions
  - Understand what happened / cahnged between different versions
  - Record reasons for changes
- Allows to work simultaneously on the same code (when using remote server)
  - Alone (on different computers)
  - Multiple people in a collaboration
- Maintain several parallel versions of the same code in a systematic way
- Many tools available (web -based services, graphical interfaces, etc.)

Different Version Control System (VCS)

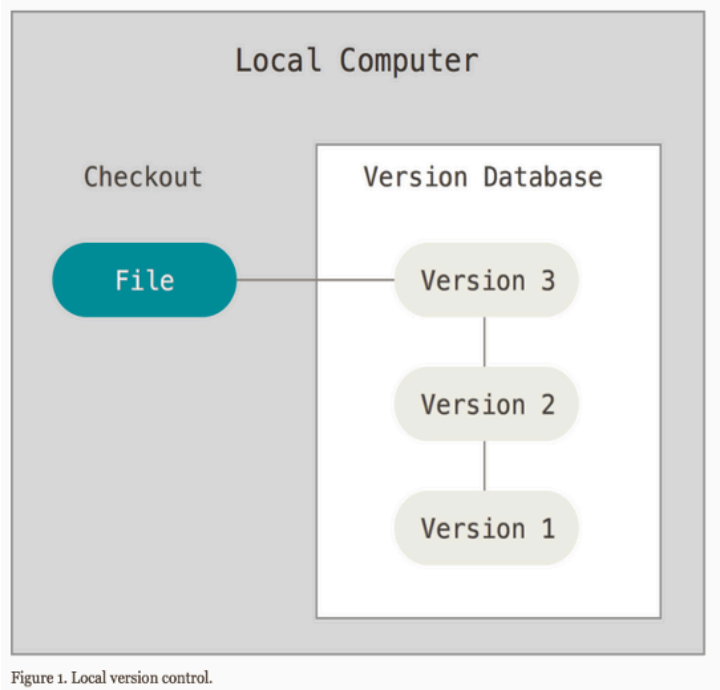
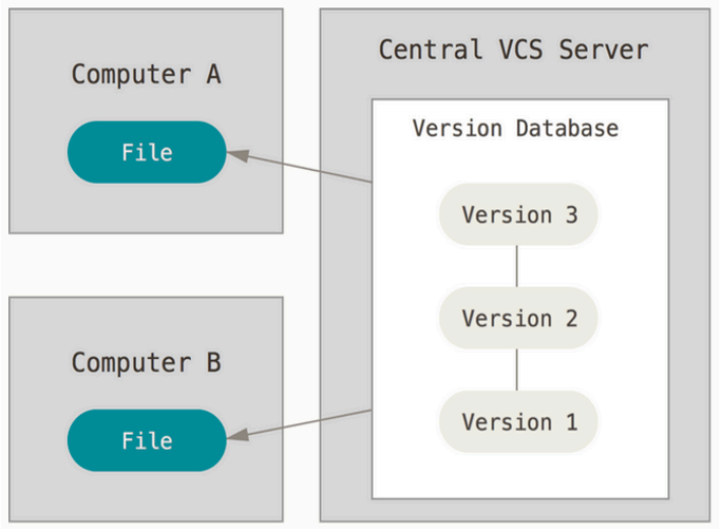
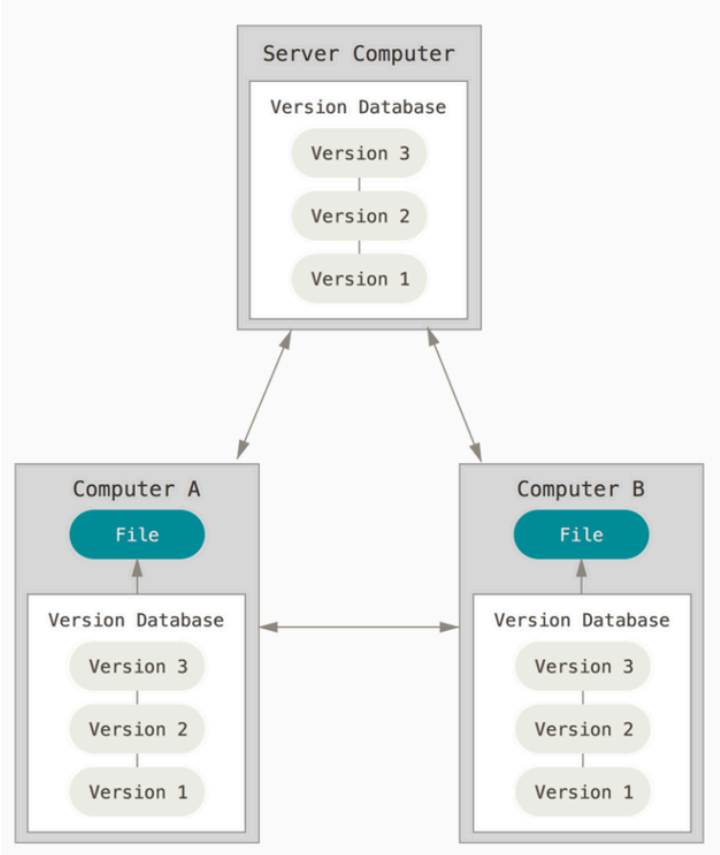


Figure 1. Local version control.

Local VCS



Centralized VCS (e.g., SVN)

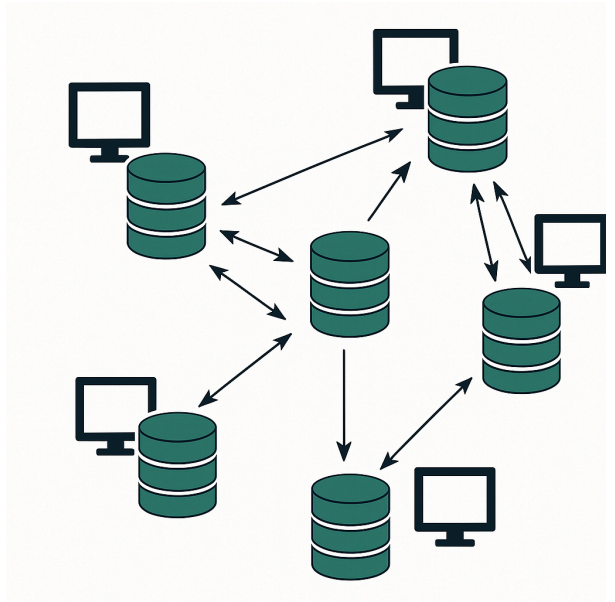


Distributed VCS (e.g., Git)

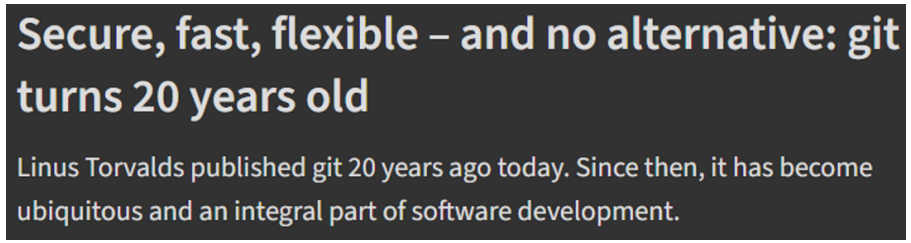
Source: book ProGit

Why Git?

- Designed for collaborative, open-source workflows
- Distributed
  - Easy and flexible code exchange
  - Back-up your code
- Widely used, fast and efficient



Happy Birthday, Git!



"I'm an egotistical bastard and name all projects after myself. First 'Linux', now 'git'", says Linus Torvalds about himself. git is British slang for "stubborn people who think they're always right and argue around".

https://www.heise.de/en/news/Secure-fast-flexible-and-no-alternative-git-turns-20-years-old-10340526.html

Practical Example: File for the Planing of a Conference

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Without versioning

conference_schedule_v0.txt
conference_schedule_v1.txt
conference_schedule_v2.txt
conference_schedule_v2_dj.txt
conference_schedule_v2_orch.txt
conference_schedule_v3.txt

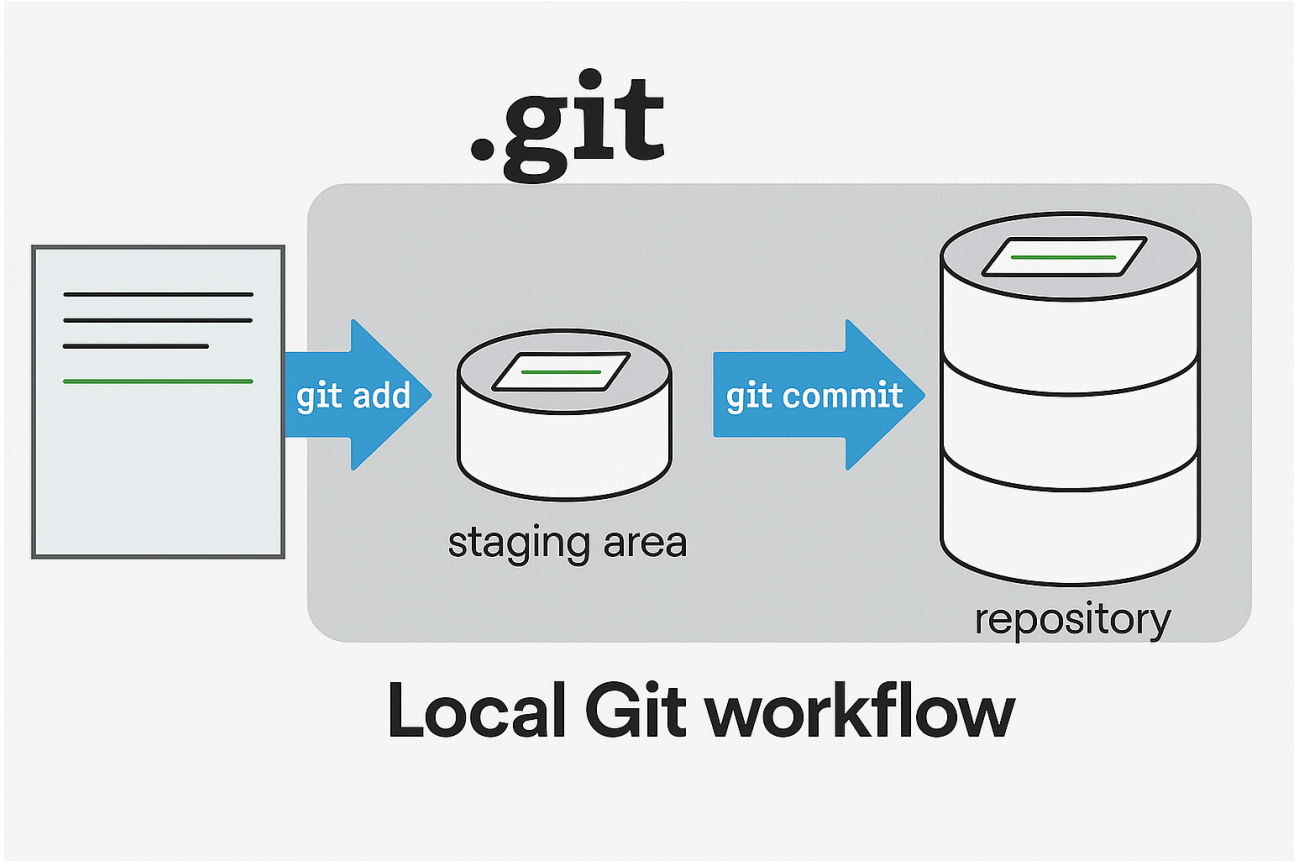
- Lots of different versions of same file
- Who remembers which one is what?

With versioning

conference_schedule.txt
.git (stores version history, date and authors of changes, etc.)

- Only one working file
- History of file documented
- Easy to combine

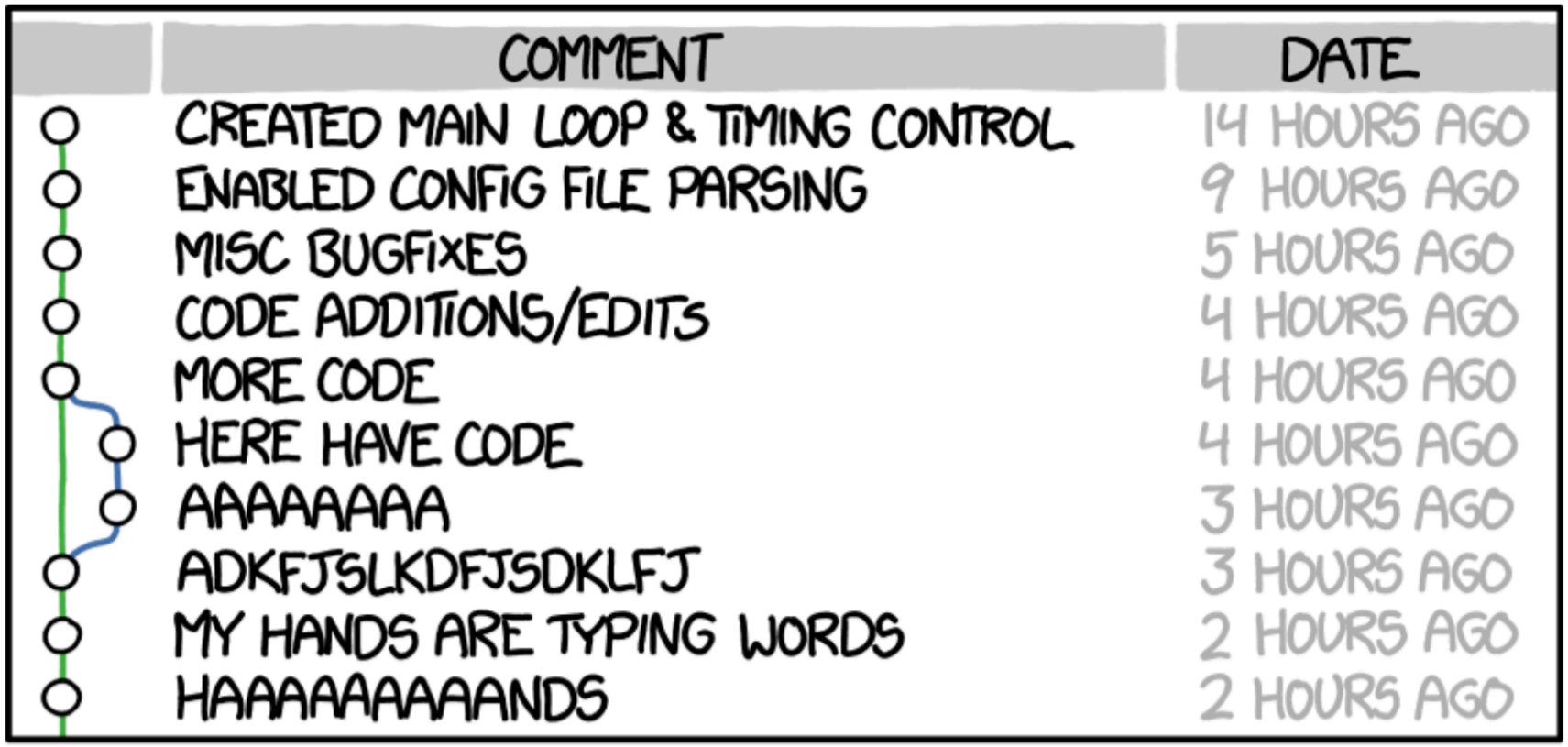
Local Git Workflow



Important Git Commands

- `git init`:
  - Creates empty Git repository
  - Creates *main* branch by default
  - Creates the *.git* folder and contents
- `git add`:
  - Saves code changes to staging area
  - Can add all or only some of the current code changes
- `git commit`:
  - Saves changes from staging area to the repository
  - Creates a unique commit ID
  - Saves a log message for the user

Commit Messages



Important Git Commands

- `git log`
  - Displays the log of all commits
- `git status`
  - Shows the status of the working copy
  - States which files have been placed in the staging area
  - Shows which files have been modified but not placed in staging area
- `git diff`
  - Shows changes between two versions of the code

Important Remarks

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It is **difficult to mess something up** with Git – don't be afraid of it!

Read what Git tells you, it is trying to help you!

Accessing Exercises

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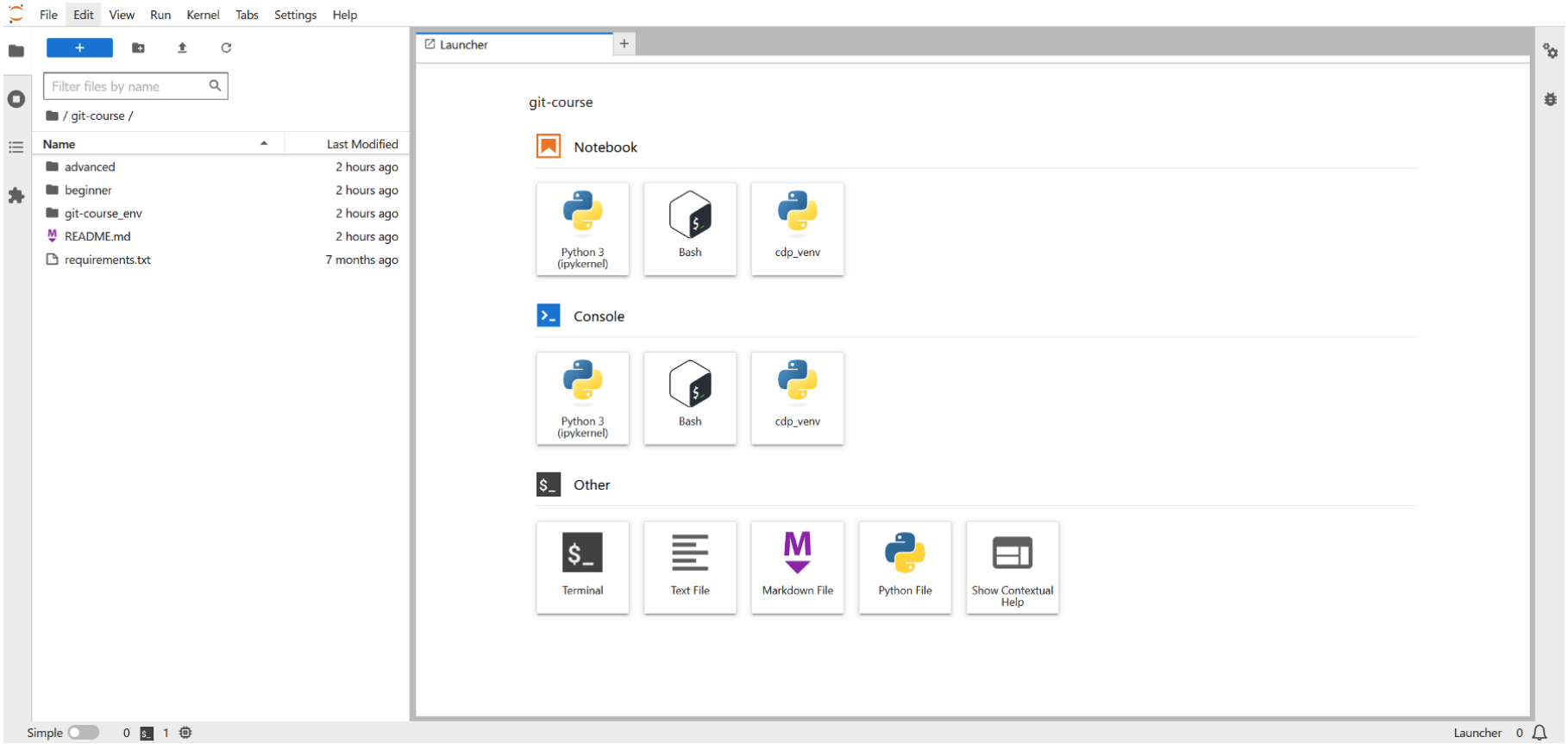
This git course is available at:

https://github.com/C2SM/git-course

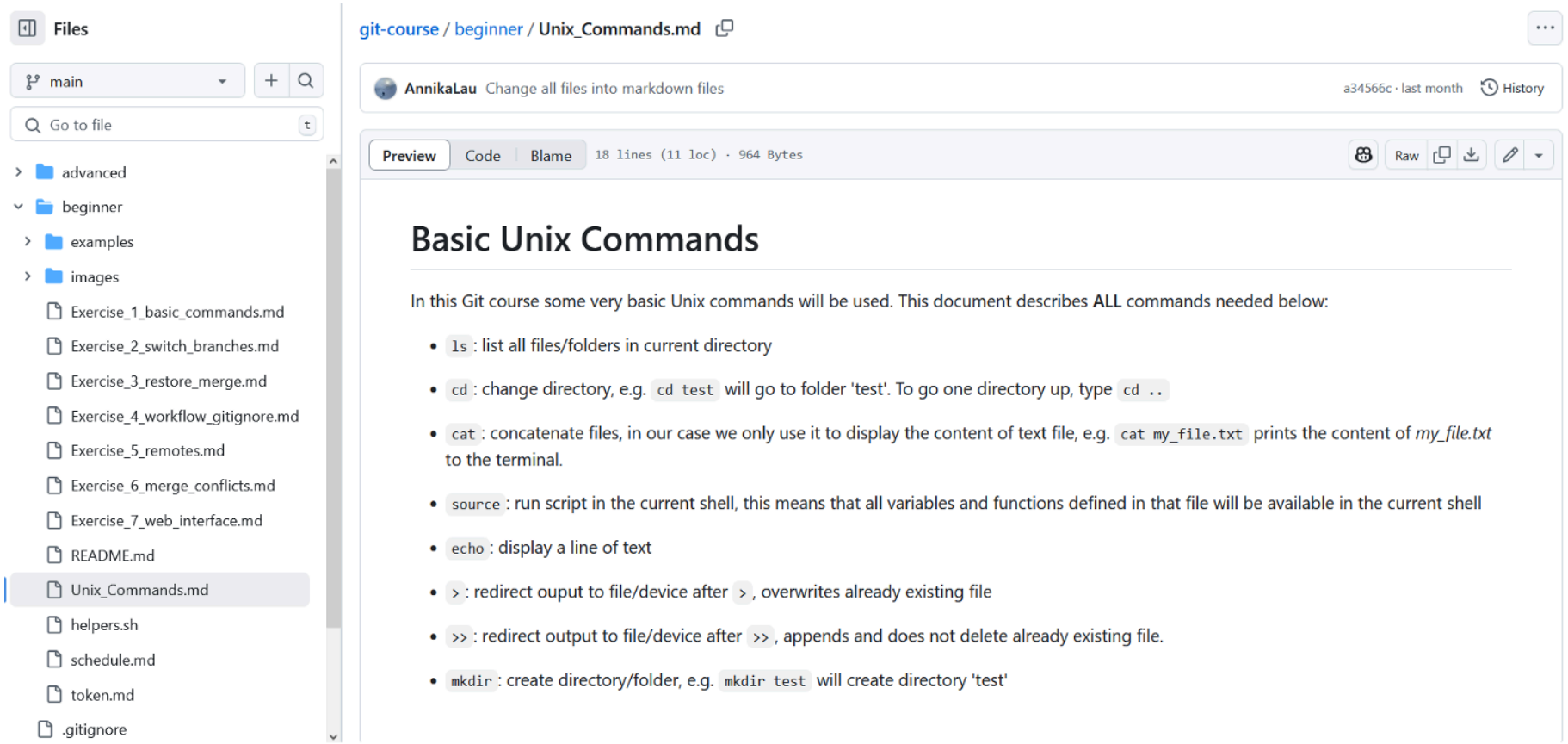
→ Select the «beginner» folder to access exercises,

e.g. «Exercise_1_basic_commands.md»

JupyterLab


File
Edit
View
Run
Kernel
Tabs
Settings
Help
Filter files by name
/ git-course /
Name
Last Modified
advanced
2 hours ago
beginner
2 hours ago
git-course_env
2 hours ago
README.md
2 hours ago
requirements.txt
7 months ago
Launcher
git-course
Notebook
Python 3 (ipykernel)
Bash
cdp_venv
Console
Python 3 (ipykernel)
Bash
cdp_venv
Other
Terminal
Text File
Markdown File
Python File
Show Contextual Help
Simple
0
1
Launcher
0

https://github.com/C2SM/git-course/blob/main/beginner/Unix_Commands.md



Exercise 1

- Create Git repository from scratch
- Track changes in files using `git add`, `git status` and `git commit`
- Know state of Git repo using `git diff` and `git log`

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Part 2:

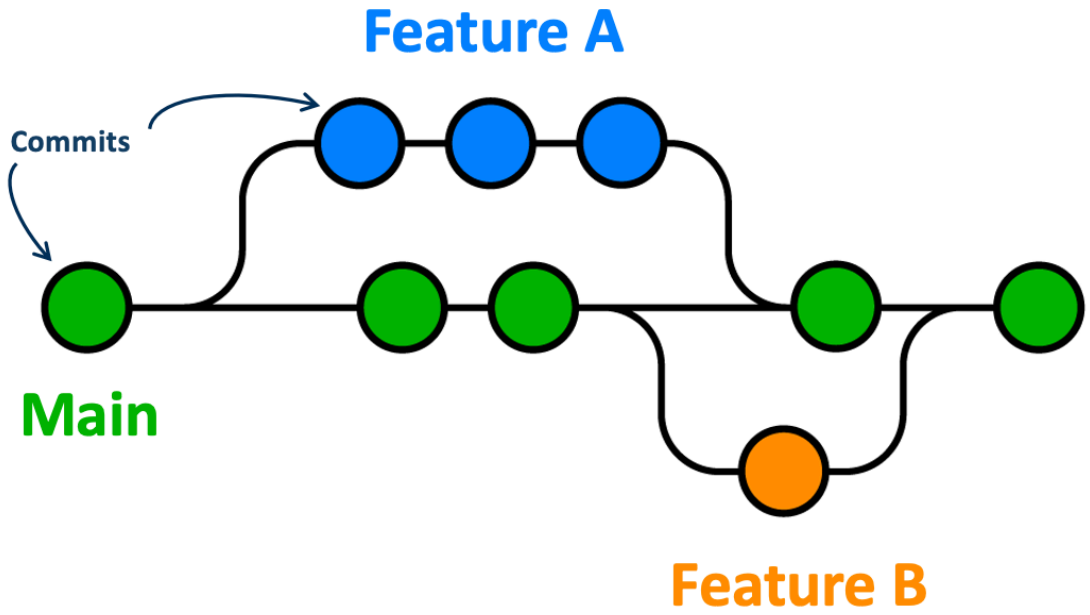
Git Branches

Remark

- Git ≥v2.23: `git switch` and `git restore`.
- Older versions: `git checkout` for both.

We will use `git switch` and `git restore` for simplicity
(`git checkout` commands in brackets)

Git Branches

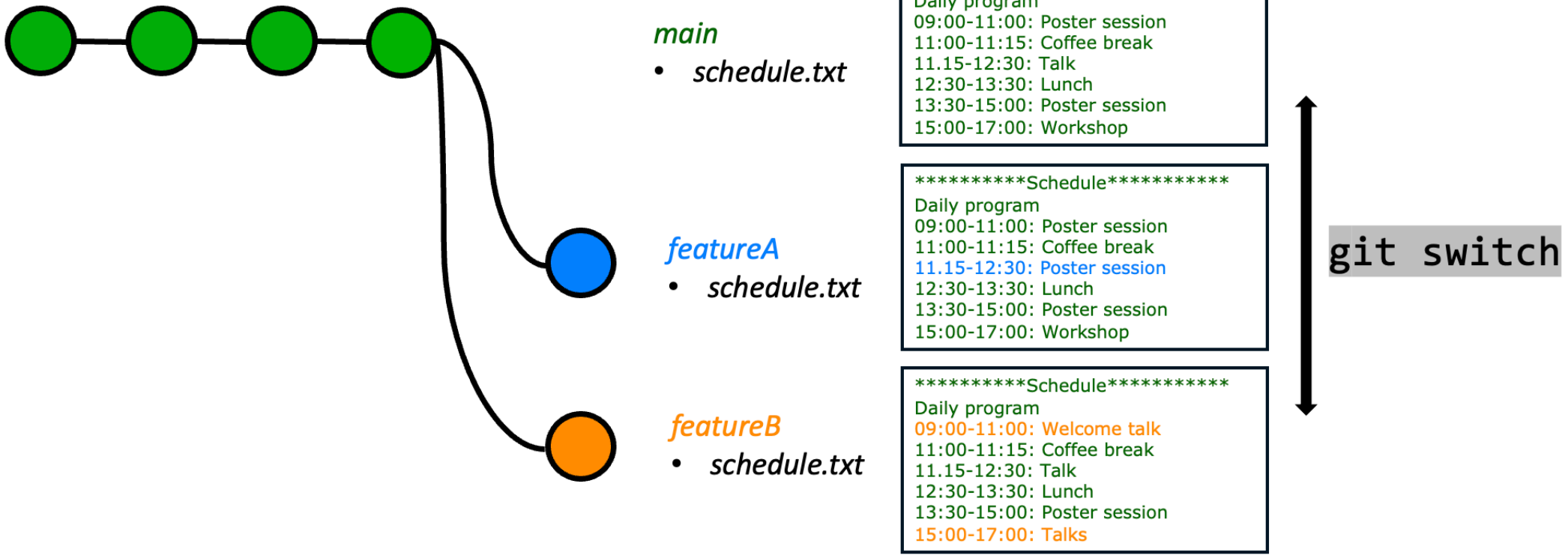


Adapted from https://blog.nobledesktop.com/learn/git/git-branches

- Multiple versions of your code
- Branch = copy of your code
- Git: create, merge and delete branches
- Advantages:
  - Easy collaboration with others
  - Experiment with new features
  - Revert changes if necessary

Git Branches

- Make changes, without affecting the main version
- Multiple brances co-exist in parallel (different versions of same file)



Exercise 2

- Work with branches
- Switch between branches using `git switch`

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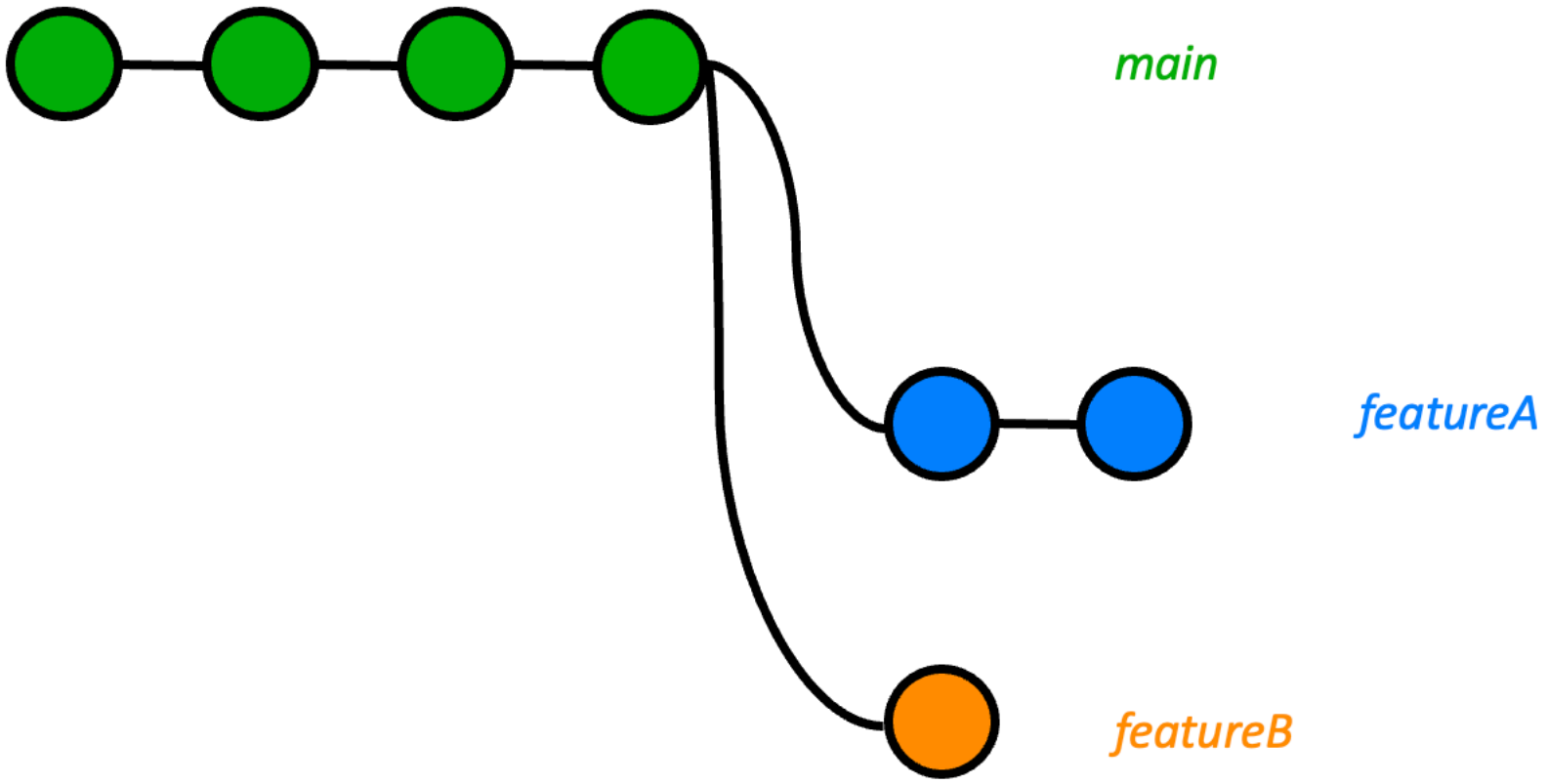
Part 3:

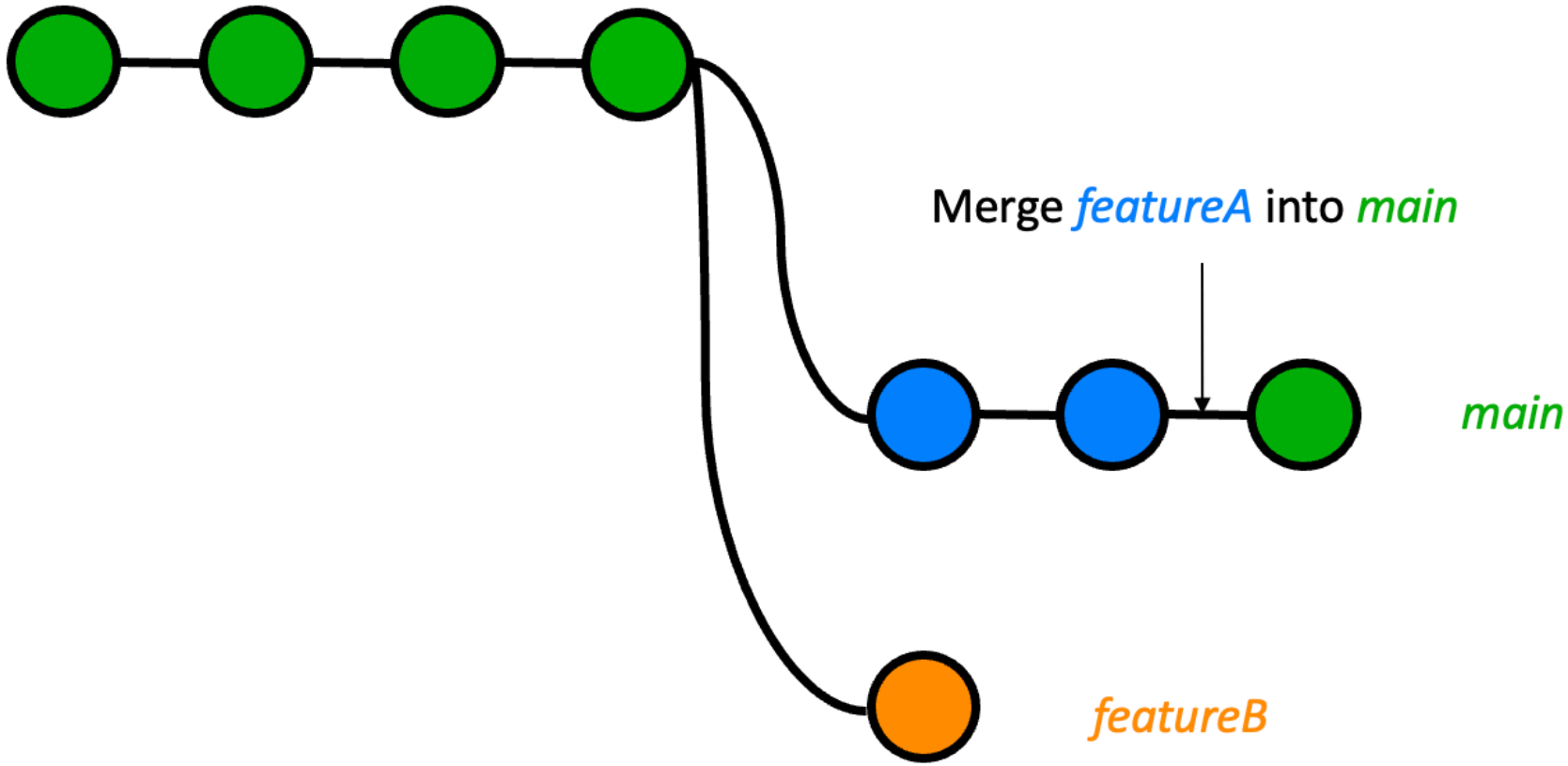
Merge, Restore and Delete

Merge Branches



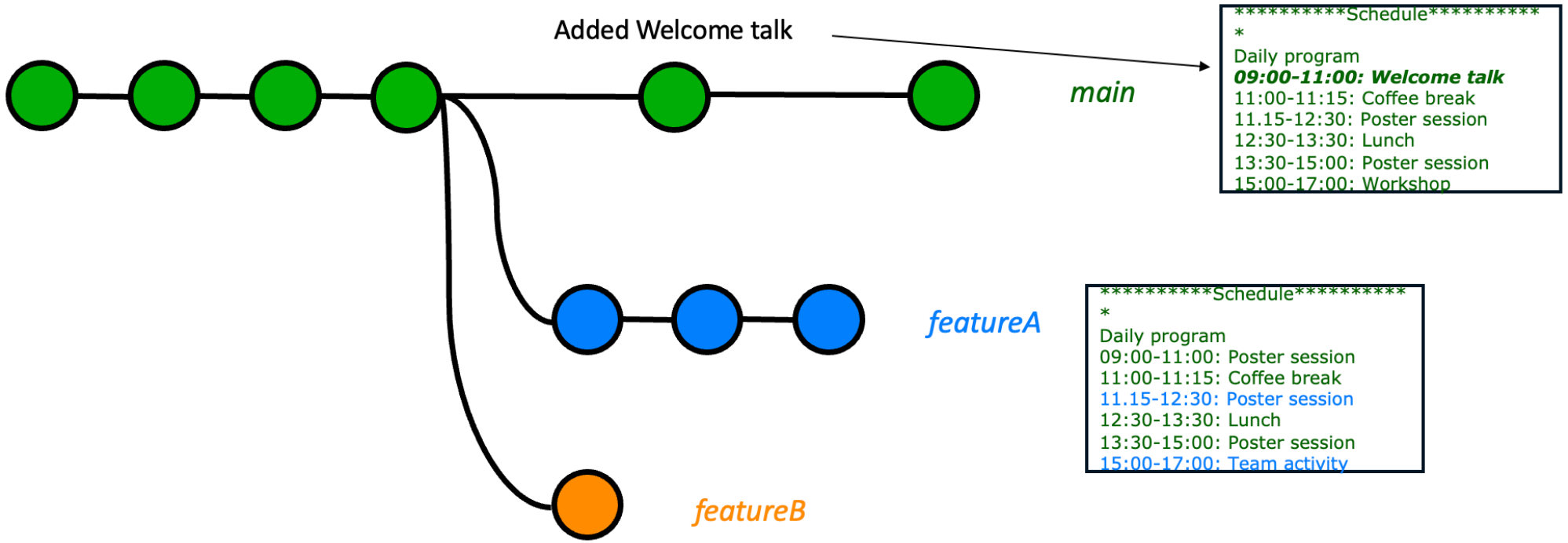
Fast-Forward Merge

- Linear history between *main* and *featureA*



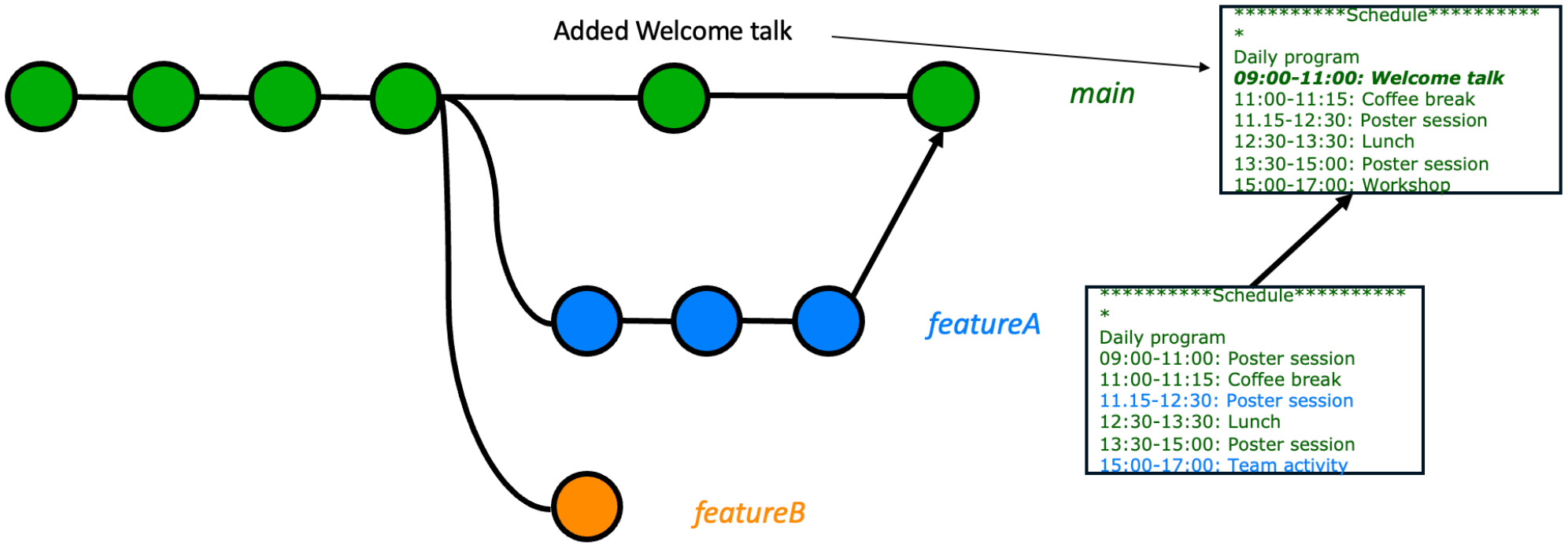
3-Way Merge

- Commits between *main* and *featureA*



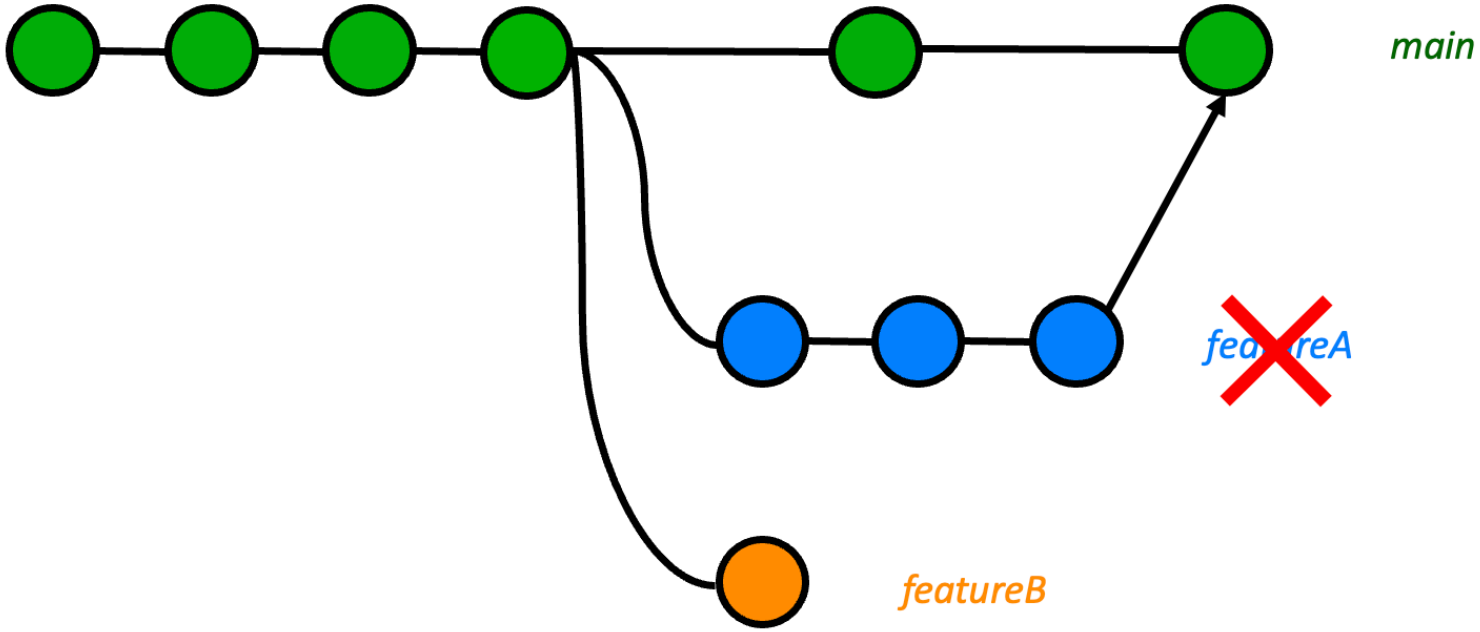
3-Way Merge

- Commits between *main* and *featureA*

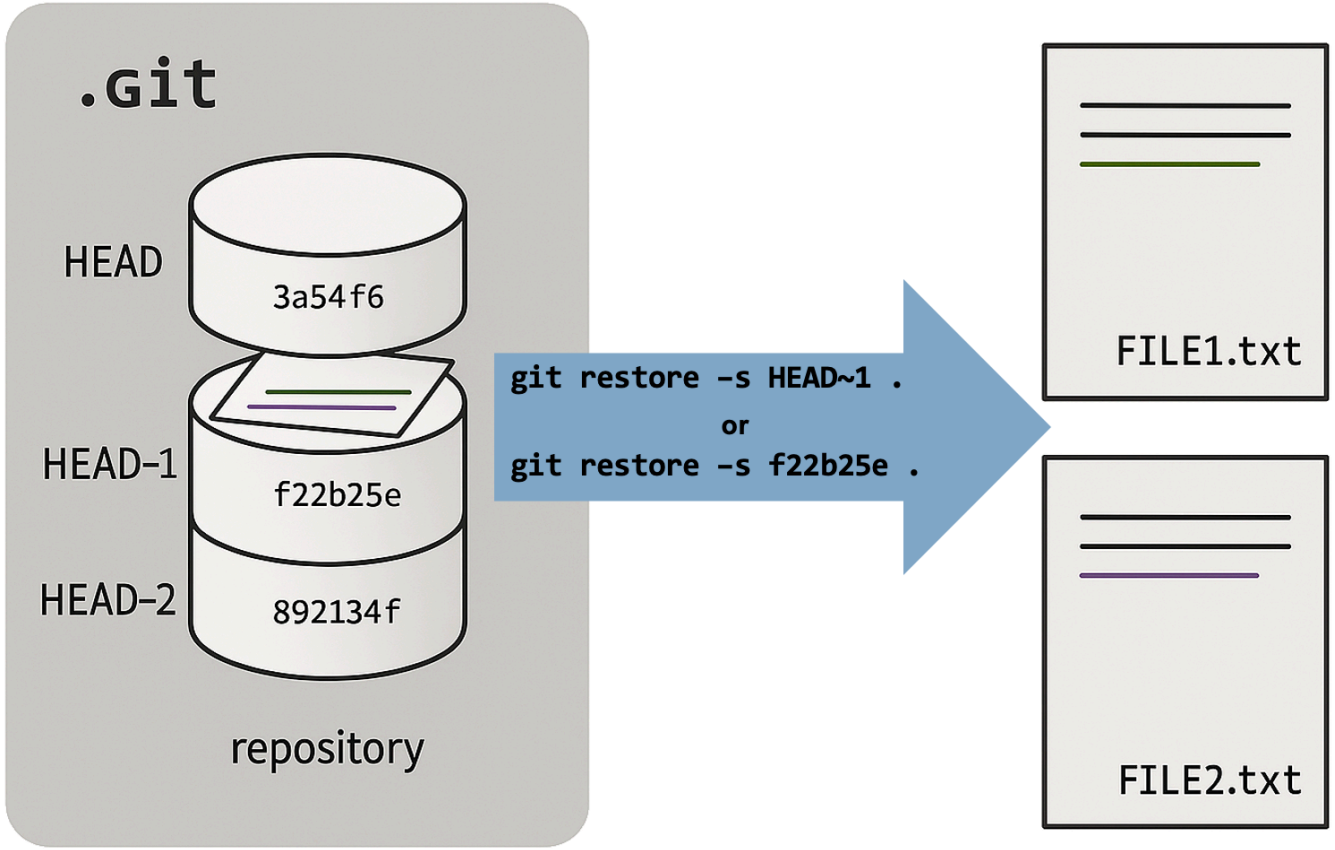


3-Way Merge

- Delete branch after merge



Git Restore


.Git
HEAD
3a54f6
HEAD-1
f22b25e
HEAD-2
892134f
repository
git restore -s HEAD~1 .
or
git restore -s f22b25e .
FILE1.txt
FILE2.txt

Important Git Commands

- `git restore`
  - Deletes unstaged changes in working directory
  - `-s <commit ID>`: Changes file to the state at specific commit ID across branches or back in history
  - Can be used for single files (`<file_x>`) or the entire repository (`.`)
- `git merge <source branch>`
  - Combines source branch with target branch (= current branch)
  - Does not create new commit unless specified or needed (depends on your config settings)
  - Called from target branch
- `git branch`
  - list branches in local repository
  - `-a`: list all branches (remotes included)
  - `-d`: delete branch (if already merged into HEAD)
  - `-D`: delete branch (force)

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Part 4:

Practice workflow and .gitignore

Understanding .gitignore

- Specifies intentionally untracked files by Git
- Helps maintaining a clean repository
- Examples: binary files, temporary files, log files, etc.
- Example .gitignore:

```
*.log        # ignore all log files
.DS_Store    # ignore macOS system files
temp/        # ignore ALL files in ANY directory named temp
```

Important Git Commands

- `git init`: creates a new repository
- `git status`: show the status of working tree (gives list of modified files)
- `git diff`: show differences between commits
- `git add`: add file to the selection of files to be committed (staging area)
- `git commit`: action to create a commit (snapshot of your project at a certain time)
- `git log`: show history of the commits
- `git switch`: switch branches
- `git restore`: restore working tree files
- `git merge`: merge changes from another branch

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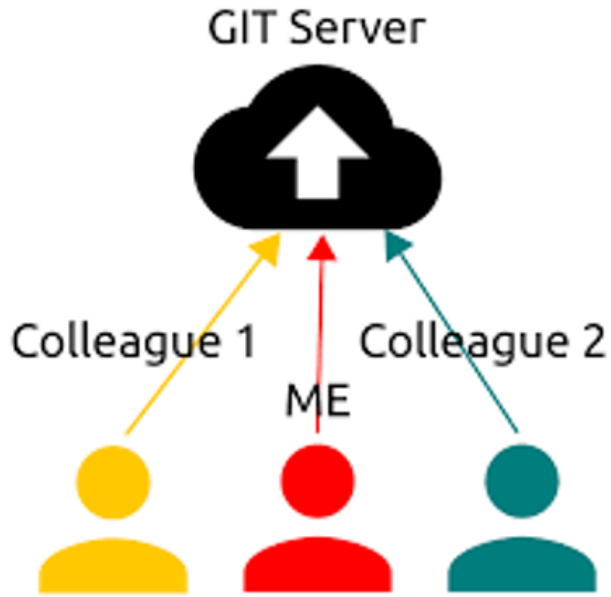
Part 5:

Remote Repositories

Remote Repositories

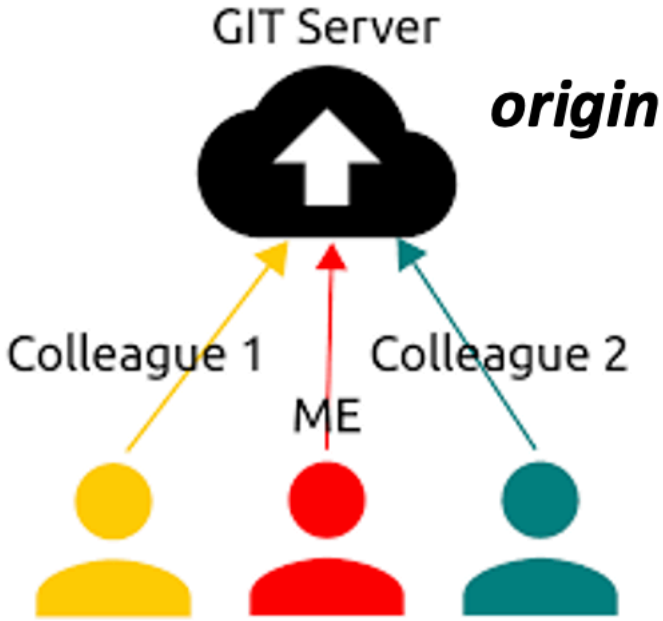
- Git repositories are usually hosted on a server (with or without web interface)
- Users clone repository to work on it locally
- The Git server is a "remote" for the local repository



Source: https://medium.com/swlh/continuous-integration-with-arduino-ce3714c19b44

Remote Repositories

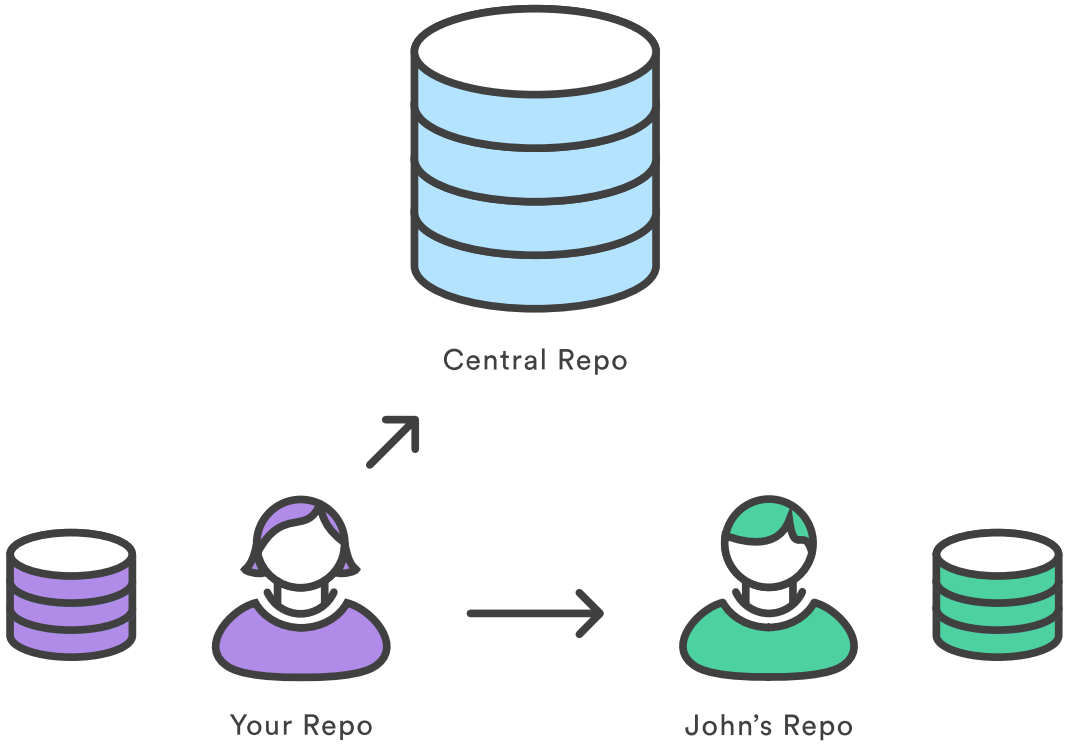
- Git automatically connects remote and local repositories when cloning
- The remote repository is called "origin" by Git



Source: https://medium.com/swlh/continuous-integration-with-arduino-ce3714c19b44

Remote Repositories

- Remotes don't have to be web-hosted
- Can have multiple remote repositories



Source: https://east.fm/refcards/git/sync/syncing.html

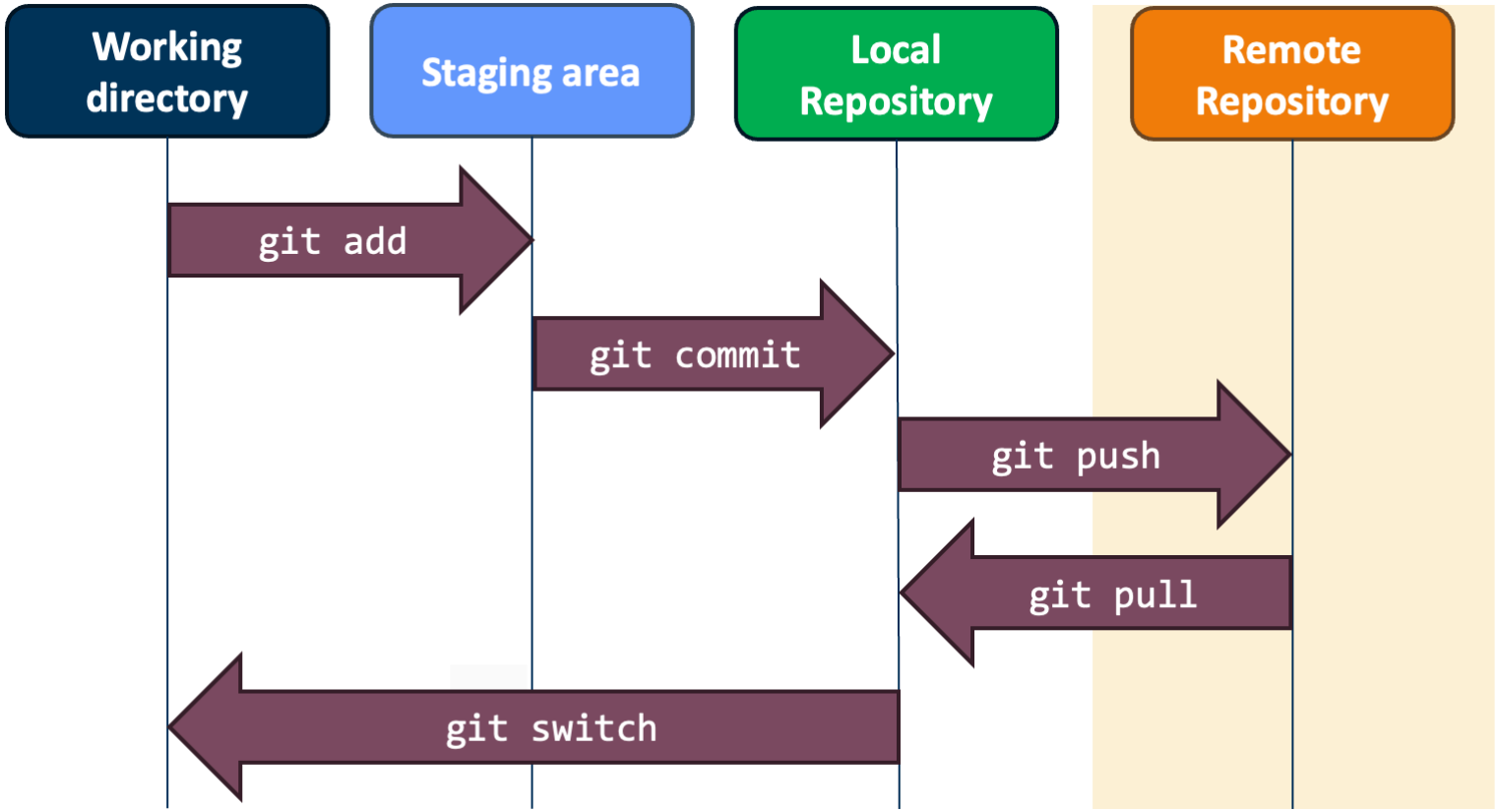
Remote Branches

- Your local repository will have local branches AND remote branches
- You can set up local branches to track the remote ones

```
# use "git branch -a" to view ALL of the branches in your local repository
git branch -a
```

```
* main
  remotes/my_remote/main
  remotes/my_remote/updated_schedules
```

Exchanging Code with Remote Repositories



```
git pull
    =
git fetch
    +
git merge
```

Modified after https://salesforcecodex.com/salesforce/useful-git-commands/

Exercise 5: Remote Repositories

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- Add a remote repository
- Examine remote branches
- Exchange information with a remote repository

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Part 6:

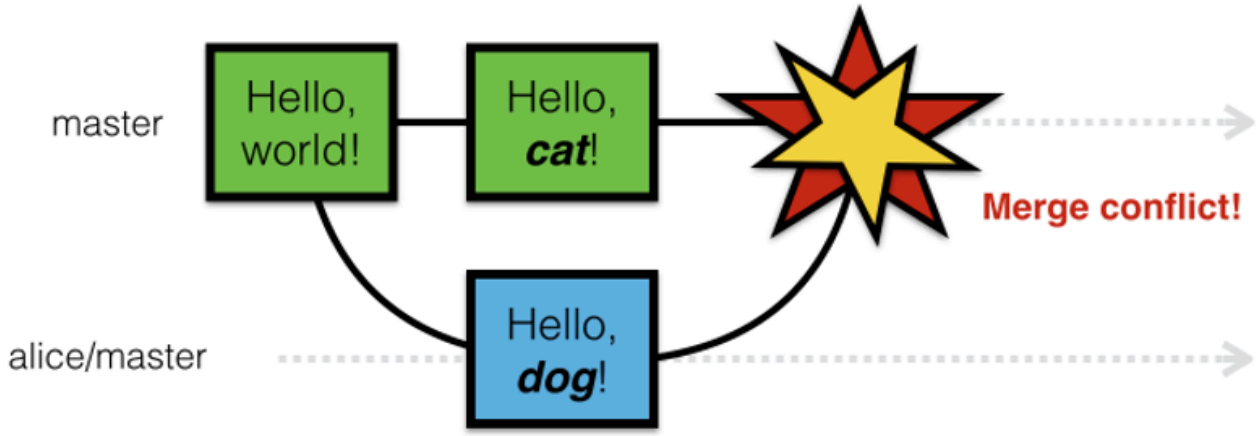
Merge conflicts

Merge conflicts

- Modifications have been made on both branches you are merging

  AND

- If you have both modified the same line in a file or one person has deleted a file another has modified



Merge conflicts

- Git will stop the merge and notify you that you have conflicts
- You must resolve these conflicts before the merge can be completed.

```
Auto-merging Hello.f90
CONFLICT (content): Merge conflict in Hello.f90
Automatic merge failed; fix conflicts and then commit the result.
silvertk@lavoisier:~/Documents/Git_tutorials/example> git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)

        both modified:   Hello.f90

no changes added to commit (use "git add" and/or "git commit -a")
```

Merge conflicts

- To resolve conficts, there are basically two options:

1. Choose the local or the remote version of a file:

   `git restore my_file --ours/theirs`

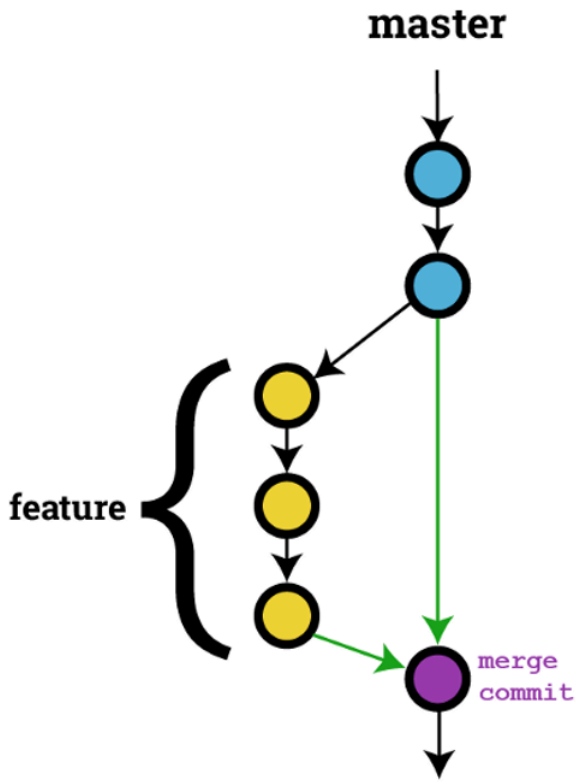
   (git checkout my_file –ours/theirs)

2. Open file, find conflict markers and decide how to solve the conflict:

```
<<<<<<< HEAD
Hello Cats!
=======
Hello Dogs!
>>>>>>> branch
```

Merge Conflicts

- Use `git add` to indicate that the file is ready for the merge
- Once all the conflicted files are resolved and added to the staging area, you use `git commit` to finalize the merge



https://haydar-ai.medium.com/learning-how-to-git-merging-branches-and-resolving-conflict-61652834d4b0

Exercise 6: Merge Conflicts

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- Learn how to deal with merge conflicts:
  - Undo merge
  - Choose preferred version
  - Adapt file directly

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Part 7:

Repository Managers

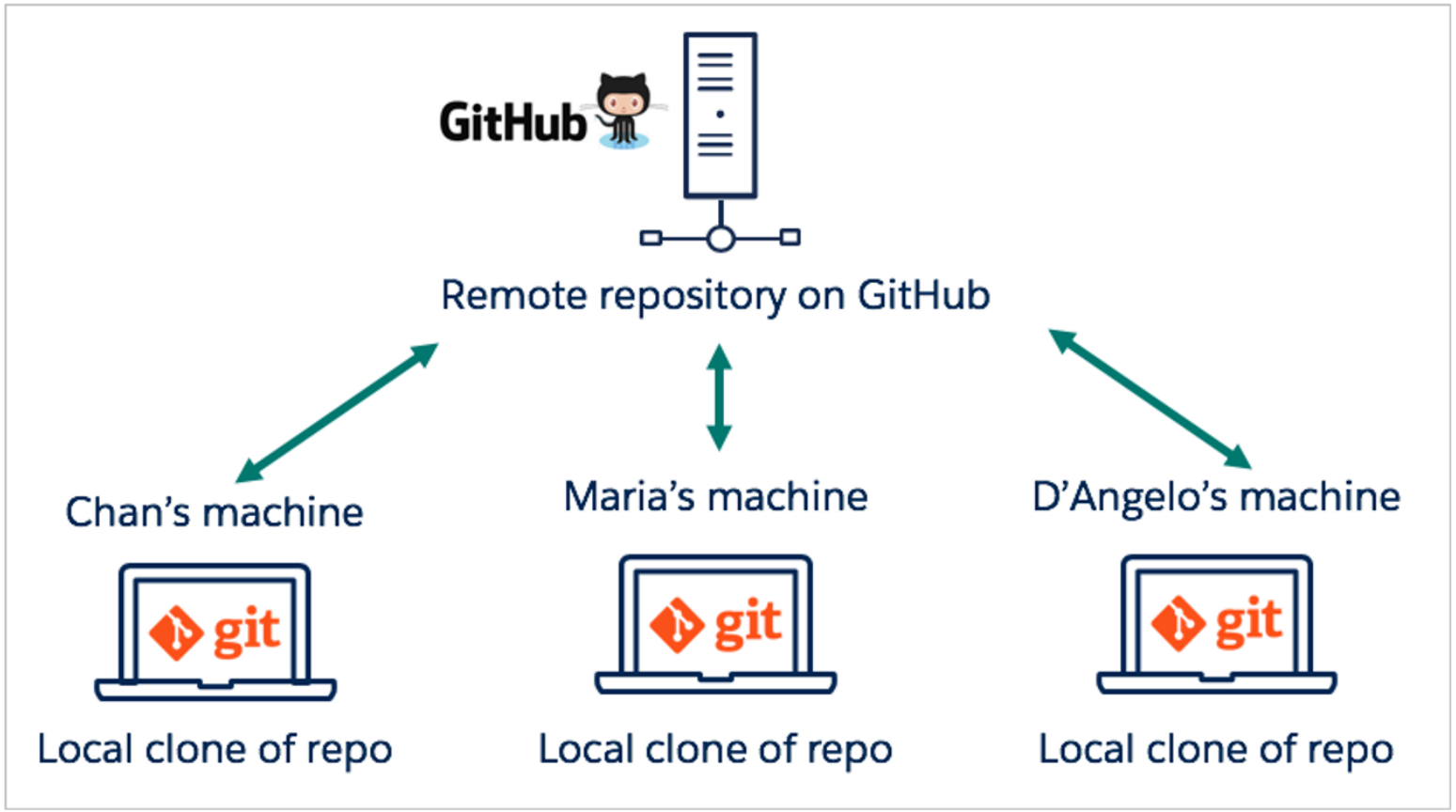
Repository Managers

- Interfaces for Git repositories
- Can use the web service or install locally
- Pricing varies – check for academic discounts!
- Examples:
  - C2SM hosts software on GitHub web interface
  - Empa and IAC use local Gitlab servers

Repository Manager Features

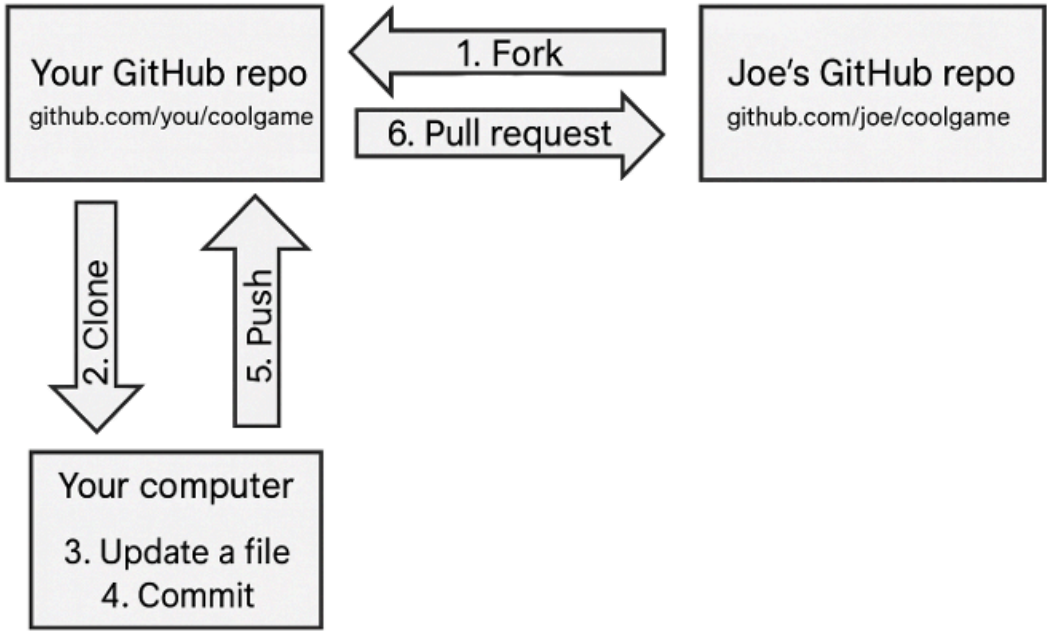
- Easily examine commits and files
- Collaboration tools (more later!)
- Issue reporting
- Code review
- Pull/merge requests
- Automation of testing

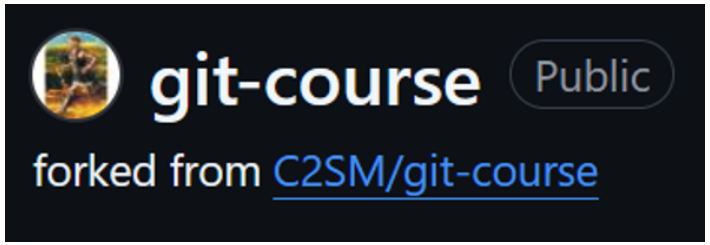
Single User or Small Group Workflow



https://blog.devgenius.io/entering-into-devops-01-git-basics-2898fbbb4b5c

Large Group Workflow: Using Forks





http://www.programmersought.com/article/4701192164/

Exercise 7: Repository Managers

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- Access code from a Git web interface
- Push code changes to a Git web interface
- Examine the code repository on a Git web interface